# Recyclable and Household Waste Classification

Era Malaj, Emi Brahaj, Aleksander Murati

Presentation Date: 16/01/2026

# CONTENTS


1 Introduction & Objectives

2 Dataset

3 Related Work

4 Intelligent Agent Design (PEAS)

5 Preprocessing

6 Models

7 Results

8 Conclusions

# 1. Introduction & Objectives

## 1 Introduction

Efficient waste sorting is a critical challenge in modern waste management systems. Incorrect disposal of household waste leads to environmental pollution, recycling contamination, and increased landfill usage. Recent advances in Artificial Intelligence enable automated waste classification systems that can assist humans in making accurate disposal decisions.
This project explores the use of machine learning and intelligent agent design to classify household waste images and recommend appropriate disposal bins.

## 2 Motivation & Objectives

The motivation of this project is to investigate how intelligent agents can support sustainable waste management through automated decision-making. Manual waste sorting is error-prone and inefficient, especially when waste categories are visually similar.

The objectives of this project are:

-To train machine learning models for multi-class waste image classification
-To compare classical statistical learning models with a neural network
-To integrate a rule-based reasoning agent for disposal recommendations

# 2. Dataset

Dataset Overview

A real-world household and recyclable waste image dataset
Images collected across 30 fine-grained waste categories
Each category corresponds to a distinct waste type (plastic bottles, cardboard, food waste)

**Dataset Structure**
Data organized in a folder-based format
Each subfolder represents a class label

Images stored in RGB format

**Dataset Size**

Total images: ~15,000
Number of classes: 30
Images per class: ~500 (approximately balanced)

**Why This Dataset?**

Reflects real-world variability in waste appearance
Supports multi-class classification
Suitable for evaluating classical machine learning models on visual data

Dataset Preparation
Images were organized into class-based folders (30 waste categories).
All images were resized to 64×64 RGB to reduce computational cost.
Pixel values were normalized to the range [0, 1].
Corrupted or unreadable images were skipped during loading.

Feature Representation
Images were flattened into numerical feature vectors.
This representation was used for classical statistical learning models.

Learning Models (Statistical Learning)
A Random Forest classifier was trained on the full training set.
A Support Vector Machine (RBF kernel) was trained on a representative subset due to computational constraints.

Decision-Making Agent (Logical Reasoning / Planning)
A rule-based reasoning module mapped predicted waste classes to appropriate disposal bins.
This simulates an intelligent agent combining perception (classification) and action (disposal decision).

Evaluation
Model performance was evaluated using accuracy, precision, recall, and F1-score.

# 3. Related Work: Previous Studies

Classification Report:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| aerosol_cans | 0.48 | 0.33 | 0.40 | 48 |
| aluminum_food_cans | 0.25 | 0.08 | 0.12 | 49 |
| aluminum_soda_cans | 0.29 | 0.26 | 0.27 | 47 |
| cardboard_boxes | 0.43 | 0.44 | 0.43 | 48 |
| cardboard_packaging | 0.31 | 0.19 | 0.23 | 48 |
| clothing | 0.36 | 0.46 | 0.40 | 48 |
| coffee_grounds | 0.68 | 0.93 | 0.79 | 44 |
| disposable_plastic_cutlery | 0.67 | 0.54 | 0.60 | 48 |
| eggshells | 0.57 | 0.60 | 0.59 | 48 |
| food_waste | 0.42 | 0.76 | 0.54 | 46 |
| glass_beverage_bottles | 0.38 | 0.51 | 0.44 | 47 |
| glass_cosmetic_containers | 0.36 | 0.44 | 0.40 | 45 |
| glass_food_jars | 0.46 | 0.53 | 0.50 | 47 |
| magazines | 0.51 | 0.73 | 0.60 | 48 |
| newspaper | 0.31 | 0.52 | 0.39 | 48 |
| office_paper | 0.30 | 0.24 | 0.27 | 46 |
| paper_cups | 0.50 | 0.18 | 0.27 | 49 |
| plastic_cup_lids | 0.24 | 0.19 | 0.21 | 47 |
| plastic_detergent_bottles | 0.66 | 0.39 | 0.49 | 49 |
| plastic_food_containers | 0.47 | 0.47 | 0.47 | 49 |
| plastic_shopping_bags | 0.40 | 0.30 | 0.34 | 47 |
| plastic_soda_bottles | 0.44 | 0.62 | 0.51 | 47 |
| plastic_straws | 0.91 | 0.68 | 0.78 | 47 |
| plastic_trash_bags | 0.59 | 0.62 | 0.61 | 48 |
| plastic_water_bottles | 0.53 | 0.65 | 0.59 | 49 |
| shoes | 0.44 | 0.23 | 0.31 | 47 |
| steel_food_cans | 0.32 | 0.40 | 0.36 | 47 |
| styrofoam_cups | 0.69 | 0.63 | 0.66 | 49 |
| styrofoam_food_containers | 0.49 | 0.63 | 0.55 | 49 |
| tea_bags | 0.39 | 0.28 | 0.33 | 50 |
| accuracy | | | 0.46 | 1429 |
| macro avg | 0.46 | 0.46 | 0.45 | 1429 |
| weighted avg | 0.46 | 0.46 | 0.45 | 1429 |

This model's architecture is based on Convolutional neural network (CNN) architecture. The model was trained and its performance was evaluated.

This code is used to evaluate a neural network (CNN / MLP) after training. Specifically, it checks:

Overfitting vs underfitting

Training vs validation performance

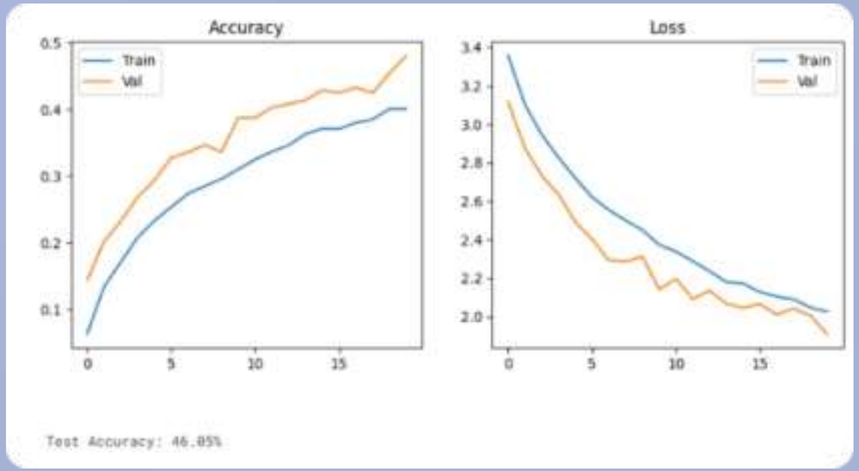
Final test accuracy

Confusion matrix

Classification report (precision / recall / F1)

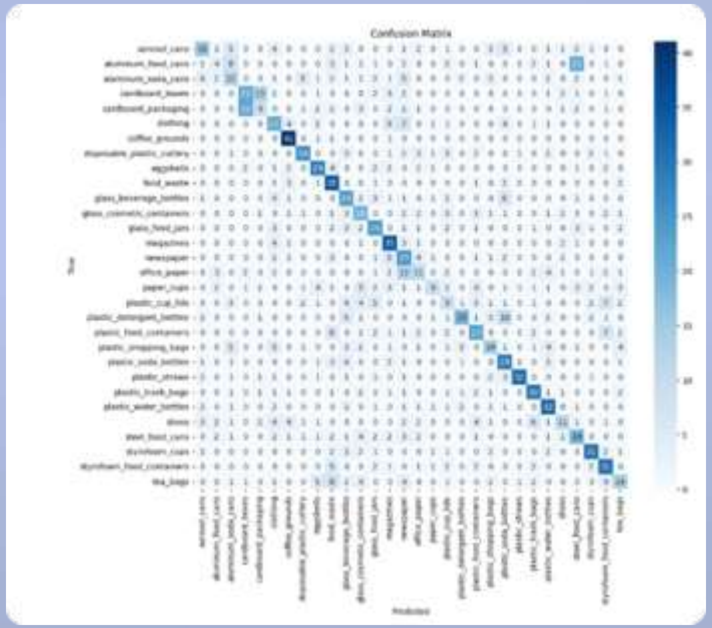
Their accuracy was 0.46.

The full code would lengthen this presentation, but can be viewed in the original users research, for the sake of time we have only included their confusion matrix and report.




https://www.kaggle.com/code/meloddy/recyclable-waste-classifier

https://www.kaggle.com/meloddy



# 4.Intelligent Agent Design (PEAS)

### Performance Measure (P)

Classification Accuracy and F1-score

Correct disposal bin recommendation

Reduction of misclassified recyclable waste

### Environment (E)

Static, real-world waste images

Partially observable (image-based perception only)

Multi-class, visually complex environment

### Actuators (A)

Disposal bin recommendation

Classification output to the user/system

### Sensors (S)

Image input captured from a camera or image dataset

# 5.Preprocessing

All images were resized to a uniform resolution of 64×64 pixels and converted to RGB format to standardize the input. Pixel values were normalized to the range [0,1] to improve numerical stability during training.

For classical machine learning models, images were flattened into one-dimensional feature vectors. For the Convolutional Neural Network, the original spatial structure of the images was preserved.

Corrupted or unreadable images were automatically skipped during data loading to ensure dataset integrity.

# 6. Models

These models allow comparison between classical machine learning and deep learning approaches for image-based waste classification.

### Model 1: Random Forest

An ensemble-based classifier that improves generalization by combining multiple decision trees

### Model 2: Decision Tree

A baseline statistical learning model used for interpretability and comparison

### Model 3: CNN

A deep learning model that learns visual features directly from raw images

### Model 4: Rule-Based Reasoning Agent

- Maps predicted class → disposal bin

Maps predicted class → disposal bin

# Random Forest Confusion Matrix

The Random Forest classifier was used as the main statistical learning model for waste image classification. By combining multiple decision trees, the model reduces overfitting and improves generalization compared to a single decision tree.

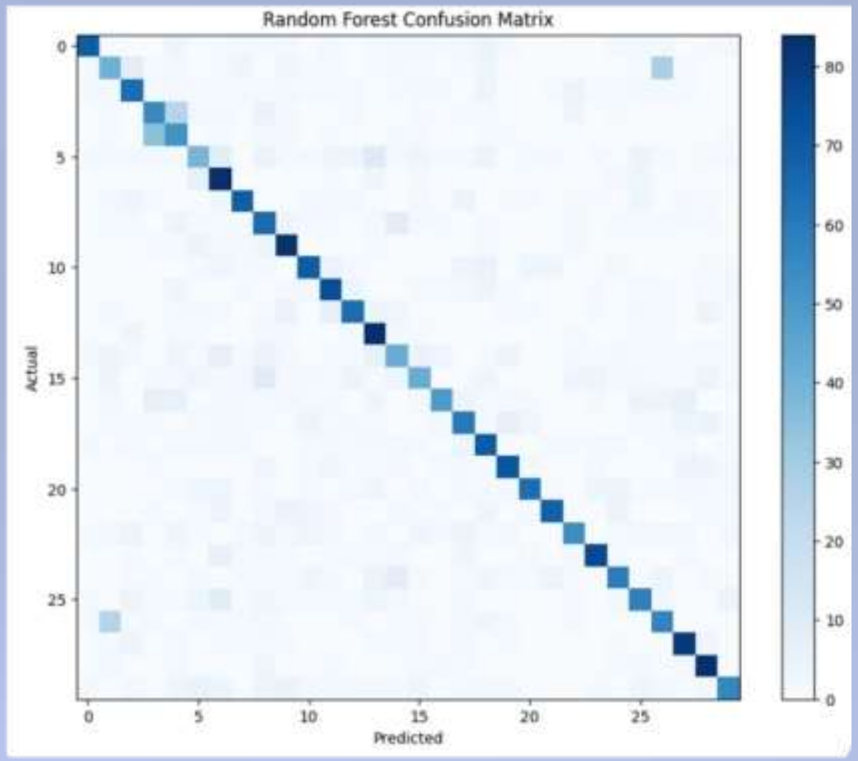
The confusion matrix shows strong diagonal dominance, indicating that a majority of samples are correctly classified across the 30 waste categories. Most misclassifications occur between visually similar classes, such as different types of plastic or paper-based waste.

Overall, the Random Forest achieved the highest accuracy among the tested models, demonstrating its robustness and suitability for high-dimensional image data when using classical machine learning techniques.

Random Forest Accuracy: 0.63

Random Forest Accuracy: 0.6343333333333333

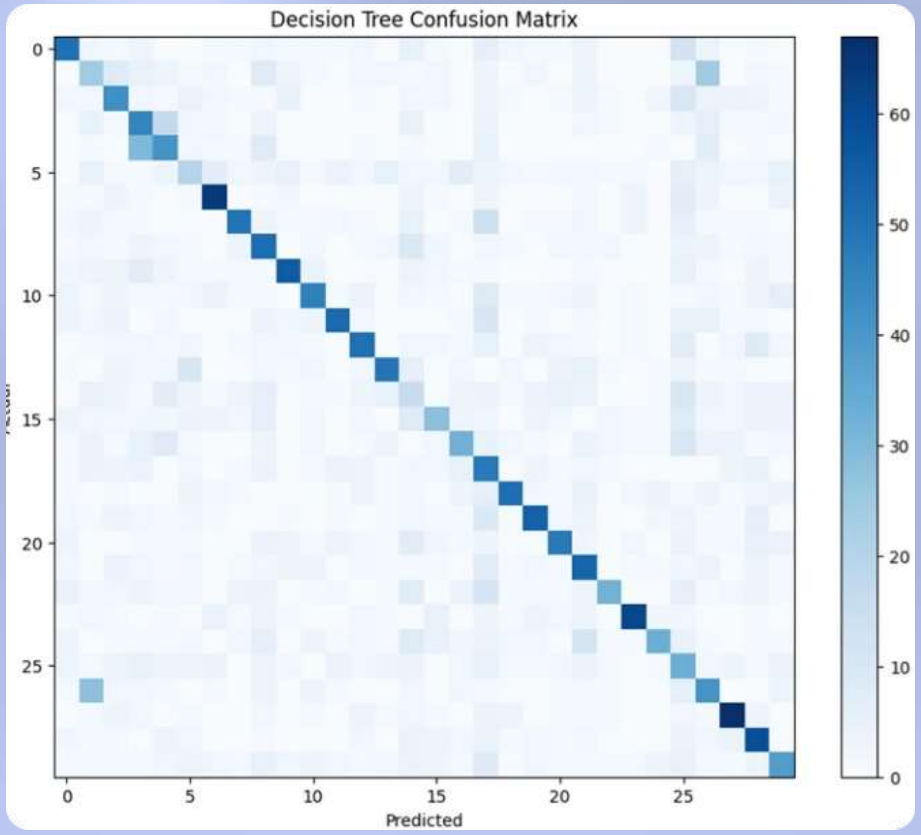
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| aerosol_cans | 0.85 | 0.69 | 0.76 | 100 |
| aluminum_food_cans | 0.45 | 0.41 | 0.43 | 100 |
| aluminum_soda_cans | 0.65 | 0.64 | 0.64 | 100 |
| cardboard_boxes | 0.50 | 0.55 | 0.52 | 100 |
| cardboard_packaging | 0.44 | 0.52 | 0.48 | 100 |
| clothing | 0.49 | 0.39 | 0.44 | 100 |
| coffee_grounds | 0.57 | 0.84 | 0.68 | 100 |
| disposable_plastic_cutlery | 0.73 | 0.69 | 0.71 | 100 |
| eggshells | 0.51 | 0.65 | 0.57 | 100 |
| food_waste | 0.65 | 0.83 | 0.73 | 100 |
| glass_beverage_bottles | 0.69 | 0.69 | 0.69 | 100 |
| glass_cosmetic_containers | 0.62 | 0.74 | 0.68 | 100 |
| glass_food_jars | 0.70 | 0.64 | 0.67 | 100 |
| magazines | 0.63 | 0.84 | 0.72 | 100 |
| newspaper | 0.48 | 0.42 | 0.45 | 100 |
| office_paper | 0.53 | 0.42 | 0.47 | 100 |
| paper_cups | 0.74 | 0.49 | 0.59 | 100 |
| plastic_cup_lids | 0.58 | 0.60 | 0.59 | 100 |
| plastic_detergent_bottles | 0.59 | 0.70 | 0.64 | 100 |
| plastic_food_containers | 0.75 | 0.72 | 0.73 | 100 |
| plastic_shopping_bags | 0.80 | 0.64 | 0.71 | 100 |
| plastic_soda_bottles | 0.86 | 0.68 | 0.76 | 100 |
| plastic_straws | 0.69 | 0.54 | 0.61 | 100 |
| plastic_trash_bags | 0.79 | 0.75 | 0.77 | 100 |
| plastic_water_bottles | 0.66 | 0.59 | 0.62 | 100 |
| shoes | 0.61 | 0.58 | 0.59 | 100 |
| steel_food_cans | 0.53 | 0.57 | 0.55 | 100 |
| styrofoam_cups | 0.74 | 0.81 | 0.77 | 100 |
| styrofoam_food_containers | 0.73 | 0.84 | 0.78 | 100 |
| tea_bags | 0.70 | 0.56 | 0.62 | 100 |
| accuracy | | | 0.63 | 3000 |
| macro avg | 0.64 | 0.63 | 0.63 | 3000 |
| weighted avg | 0.64 | 0.63 | 0.63 | 3000 |



# Decision Tree Confusion Matrix

Decision Tree Accuracy: 0.44433333333333336

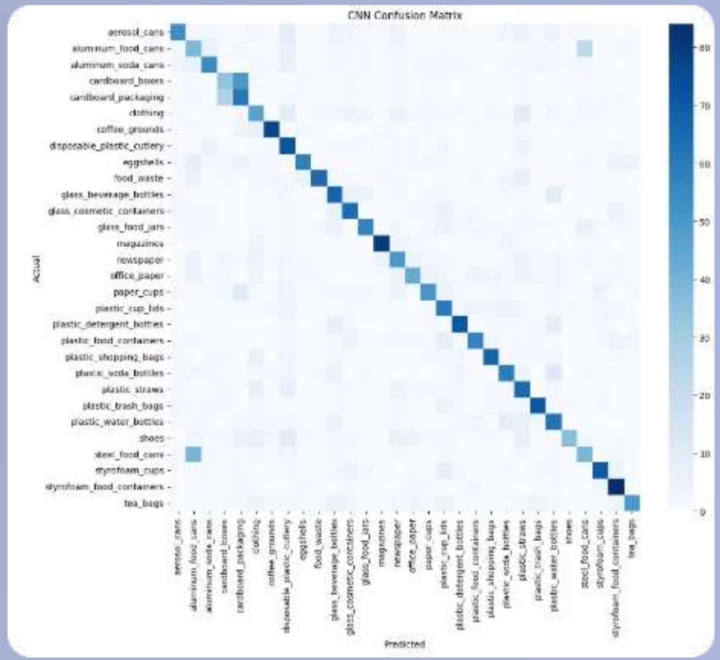
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| aerosol_cans | 0.53 | 0.50 | 0.52 | 100 |
| aluminum_food_cans | 0.23 | 0.24 | 0.24 | 100 |
| aluminum_soda_cans | 0.40 | 0.42 | 0.41 | 100 |
| cardboard_boxes | 0.34 | 0.45 | 0.38 | 100 |
| cardboard_packaging | 0.38 | 0.41 | 0.39 | 100 |
| clothing | 0.26 | 0.20 | 0.23 | 100 |
| coffee_grounds | 0.63 | 0.64 | 0.63 | 100 |
| disposable_plastic_cutlery | 0.65 | 0.49 | 0.56 | 100 |
| eggshells | 0.36 | 0.51 | 0.42 | 100 |
| food_waste | 0.62 | 0.56 | 0.59 | 100 |
| glass_beverage_bottles | 0.55 | 0.46 | 0.50 | 100 |
| glass_cosmetic_containers | 0.65 | 0.52 | 0.58 | 100 |
| glass_food_jars | 0.52 | 0.50 | 0.51 | 100 |
| magazines | 0.61 | 0.49 | 0.54 | 100 |
| newspaper | 0.14 | 0.16 | 0.15 | 100 |
| office_paper | 0.41 | 0.28 | 0.33 | 100 |
| paper_cups | 0.43 | 0.32 | 0.37 | 100 |
| plastic_cup_lids | 0.25 | 0.48 | 0.33 | 100 |
| plastic_detergent_bottles | 0.70 | 0.51 | 0.59 | 100 |
| plastic_food_containers | 0.60 | 0.54 | 0.57 | 100 |
| plastic_shopping_bags | 0.57 | 0.48 | 0.52 | 100 |
| plastic_soda_bottles | 0.41 | 0.53 | 0.46 | 100 |
| plastic_straws | 0.67 | 0.32 | 0.43 | 100 |
| plastic_trash_bags | 0.76 | 0.61 | 0.68 | 100 |
| plastic_water_bottles | 0.62 | 0.33 | 0.43 | 100 |
| shoes | 0.19 | 0.33 | 0.24 | 100 |
| steel_food_cans | 0.32 | 0.41 | 0.36 | 100 |
| styrofoam_cups | 0.64 | 0.67 | 0.65 | 100 |
| styrofoam_food_containers | 0.46 | 0.59 | 0.52 | 100 |
| tea_bags | 0.43 | 0.38 | 0.40 | 100 |
| accuracy | | | 0.44 | 3000 |
| macro avg | 0.48 | 0.44 | 0.45 | 3000 |
| weighted avg | 0.48 | 0.44 | 0.45 | 3000 |




The Decision Tree classifier was used as a baseline statistical learning model to evaluate the performance of a single-tree approach. Decision Trees are simple and interpretable models that split the data based on feature thresholds to make predictions. The Decision Tree confusion matrix shows a clear diagonal, indicating that the model successfully learns class-specific patterns. However, compared to the Random Forest, the matrix exhibits more off-diagonal misclassifications, reflecting weaker generalization and a higher tendency to overfit. This behavior is expected for a single-tree model and highlights the performance improvements achieved by ensemble methods such as Random Forest. Despite its lower accuracy, the Decision Tree provides a useful baseline and helps highlight the performance improvement gained through ensemble learning.

Decision Tree Accuracy: 0.44

# CNN MODEL




```
Epoch 1/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 6s 17ms/step - accuracy: 0.1483 - loss: 3.0345 - val_accuracy: 0.2387 - val_loss: 2.6500
Epoch 2/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.3298 - loss: 2.3746 - val_accuracy: 0.3500 - val_loss: 2.2934
Epoch 3/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.4441 - loss: 1.9568 - val_accuracy: 0.4283 - val_loss: 2.0730
Epoch 4/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.5448 - loss: 1.5937 - val_accuracy: 0.4921 - val_loss: 1.8578
Epoch 5/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.6425 - loss: 1.2356 - val_accuracy: 0.5142 - val_loss: 1.8890
Epoch 6/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.7272 - loss: 0.9300 - val_accuracy: 0.5529 - val_loss: 1.8669
Epoch 7/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.8004 - loss: 0.6756 - val_accuracy: 0.5733 - val_loss: 1.9235
Epoch 8/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.8421 - loss: 0.5209 - val_accuracy: 0.5833 - val_loss: 2.1830
Epoch 9/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.8859 - loss: 0.3671 - val_accuracy: 0.6046 - val_loss: 2.3009
Epoch 10/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9036 - loss: 0.3200 - val_accuracy: 0.5883 - val_loss: 2.4030
Epoch 11/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9159 - loss: 0.2759 - val_accuracy: 0.6025 - val_loss: 2.6227
Epoch 12/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9335 - loss: 0.2163 - val_accuracy: 0.6071 - val_loss: 2.7663
Epoch 13/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9468 - loss: 0.1799 - val_accuracy: 0.5962 - val_loss: 2.8854
Epoch 14/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 6s 20ms/step - accuracy: 0.9484 - loss: 0.1743 - val_accuracy: 0.5867 - val_loss: 2.8695
Epoch 15/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.9505 - loss: 0.1631 - val_accuracy: 0.5917 - val_loss: 3.0973
Epoch 16/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9428 - loss: 0.1817 - val_accuracy: 0.5954 - val_loss: 2.9687
Epoch 17/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 18ms/step - accuracy: 0.9373 - loss: 0.2054 - val_accuracy: 0.5938 - val_loss: 2.9955
Epoch 18/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9483 - loss: 0.1609 - val_accuracy: 0.6075 - val_loss: 3.3203
Epoch 19/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9597 - loss: 0.1356 - val_accuracy: 0.6208 - val_loss: 3.0050
Epoch 20/20
300/300 ━━━━━━━━━━━━━━━━━━━━ 5s 17ms/step - accuracy: 0.9574 - loss: 0.1367 - val_accuracy: 0.6037 - val_loss: 3.2000
```

CNN Classification Report:

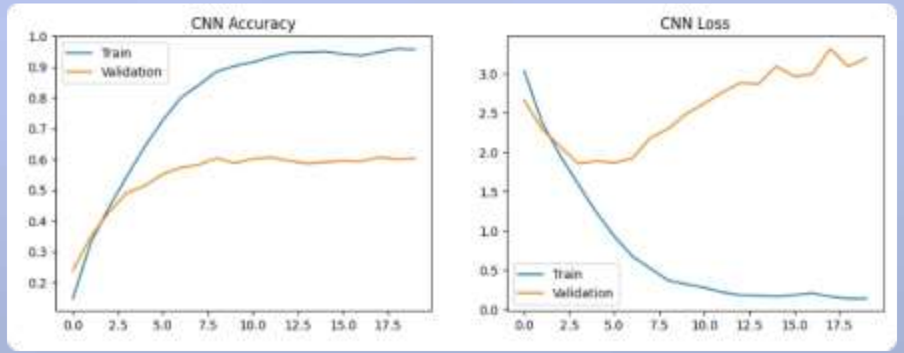
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| aerosol_cans | 0.71 | 0.54 | 0.61 | 100 |
| aluminum_food_cans | 0.29 | 0.38 | 0.33 | 100 |
| aluminum_soda_cans | 0.61 | 0.55 | 0.58 | 100 |
| cardboard_boxes | 0.41 | 0.34 | 0.37 | 100 |
| cardboard_packaging | 0.41 | 0.61 | 0.49 | 100 |
| clothing | 0.40 | 0.46 | 0.43 | 100 |
| coffee_grounds | 0.81 | 0.77 | 0.79 | 100 |
| disposable_plastic_cutlery | 0.45 | 0.72 | 0.56 | 100 |
| eggshells | 0.78 | 0.58 | 0.67 | 100 |
| food_waste | 0.86 | 0.65 | 0.74 | 100 |
| glass_beverage_bottles | 0.52 | 0.67 | 0.59 | 100 |
| glass_cosmetic_containers | 0.59 | 0.64 | 0.62 | 100 |
| glass_food_jars | 0.74 | 0.57 | 0.64 | 100 |
| magazines | 0.88 | 0.80 | 0.84 | 100 |
| newspaper | 0.48 | 0.50 | 0.49 | 100 |
| office_paper | 0.60 | 0.43 | 0.50 | 100 |
| paper_cups | 0.55 | 0.51 | 0.53 | 100 |
| plastic_cup_lids | 0.48 | 0.60 | 0.53 | 100 |
| plastic_detergent_bottles | 0.73 | 0.71 | 0.72 | 100 |
| plastic_food_containers | 0.70 | 0.57 | 0.63 | 100 |
| plastic_shopping_bags | 0.73 | 0.67 | 0.70 | 100 |
| plastic_soda_bottles | 0.61 | 0.59 | 0.60 | 100 |
| plastic_straws | 0.44 | 0.65 | 0.53 | 100 |
| plastic_trash_bags | 0.72 | 0.71 | 0.71 | 100 |
| plastic_water_bottles | 0.49 | 0.62 | 0.55 | 100 |
| shoes | 0.77 | 0.36 | 0.49 | 100 |
| steel_food_cans | 0.44 | 0.39 | 0.41 | 100 |
| styrofoam_cups | 0.74 | 0.71 | 0.72 | 100 |
| styrofoam_food_containers | 0.69 | 0.84 | 0.76 | 100 |
| tea_bags | 0.82 | 0.50 | 0.62 | 100 |
| accuracy | | | 0.59 | 3000 |
| macro avg | 0.62 | 0.59 | 0.59 | 3000 |
| weighted avg | 0.62 | 0.59 | 0.59 | 3000 |

The Convolutional Neural Network was trained on approximately 15,000 waste images belonging to 30 different categories. The model learned to extract visual features directly from raw images using multiple convolutional and pooling layers, followed by fully connected layers for classification.

During training, the model achieved a high training accuracy of approximately 96%, while the validation and test accuracy stabilized around 60%. The accuracy and loss curves indicate a clear gap between training and validation performance, suggesting overfitting.

The confusion matrix shows a strong diagonal trend, meaning that many classes are correctly identified. Most misclassifications occur between visually similar materials, such as different types of plastic, paper, and glass. Classes with distinctive appearances achieve higher precision and recall.

Overall, the CNN demonstrates the effectiveness of neural networks for image-based waste classification and serves as a strong perception component of the intelligent agent.



# Rule-Based Decision Agent – Disposal Recommendation

Agent Output Examples

Magazines → Paper bin

Cardboard boxes → Paper bin

Plastic food containers → Plastic bin

Plastic detergent bottles → Plastic bin

Glass beverage bottles → Glass bin

Steel food cans → Metal bin

```
Predicted class: magazines                   → Disposal bin: Paper
Predicted class: plastic_food_containers     → Disposal bin: Plastic
Predicted class: plastic_detergent_bottles   → Disposal bin: Plastic
Predicted class: cardboard_boxes             → Disposal bin: Paper
Predicted class: glass_beverage_bottles      → Disposal bin: Glass
Predicted class: plastic_soda_bottles        → Disposal bin: Plastic
Predicted class: plastic_trash_bags          → Disposal bin: Plastic
Predicted class: steel_food_cans             → Disposal bin: Metal
Predicted class: plastic_cup_lids            → Disposal bin: Plastic
Predicted class: plastic_soda_bottles        → Disposal bin: Plastic
```

Decision Logic

The agent uses a rule-based reasoning system

Predicted waste classes are deterministically mapped to disposal bins

Rules are based on material type (plastic, paper, glass, metal, organic)

# 7. Results

## Performance Comparison

| | |
|---|---|
| Random Forest | Accuracy: ~0.63 |
| CNN Test | Accuracy: ~0.60 |
| Decision Tree | Accuracy: ~0.44 |

# 8. Conclusions

This project demonstrated how an intelligent agent can combine perception and reasoning to support automated waste disposal decisions. Classical machine learning models provided strong baselines, while the CNN enabled end-to-end visual learning.

The rule-based decision agent successfully mapped predicted waste categories to disposal actions, simulating rational agent behavior. Future work may include data augmentation, reinforcement learning, or deployment on real-world embedded systems.

# Thank You!